

AI ASSISTED CODING

LAB ASSIGNMENT – 12.1

Name: Bitla Sairaj

Roll Number: 2503a52162

Branch: Cseai&ml

Batch: 06

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Task 1

Stack Implementation with AI Assistance

1. Objective

To design and implement a Stack data structure in Python with the help of AI assistance, supporting push, pop, peek and is_empty operations, along with proper handling of stack underflow.

2. AI Prompt Used

Act as an experienced Python developer.

Create a Python Stack data structure using a class.

Requirements:

1. Create a class named Stack.
2. Implement these methods:
 - push(element)
 - pop()
 - peek()
 - is_empty()
3. Store stack elements using an appropriate Python data structure.
4. Handle stack underflow properly when pop() or peek() is called on an empty stack.
5. Raise an appropriate exception with a clear error message.
6. Add clear docstrings to the class and every method.
7. Add comments explaining important parts of the code.
8. Include a small demonstration program that tests all methods.
9. Explain the time complexity of each operation.
10. Keep the implementation beginner-friendly.

3. Python Code

```
class Stack:
    """A simple Stack implementation using a Python list."""

    def __init__(self):
        """Initialize an empty stack."""
        self.items = []

    def push(self, element):
        """Add an element to the top of the stack."""
        self.items.append(element)

    def pop(self):
        """Remove and return the top element.

        Raises:
            IndexError: If the stack is empty.
        """
        if self.is_empty():
            raise IndexError("Stack underflow: cannot pop from an empty stack.")
        return self.items.pop()
```

```

def peek(self):
    """Return the top element without removing it.

    Raises:
        IndexError: If the stack is empty.
    """
    if self.is_empty():
        raise IndexError("Stack underflow: cannot peek at an empty stack.")
    return self.items[-1]

def is_empty(self):
    """Return True if the stack contains no elements."""
    return len(self.items) == 0

# Demonstration
stack = Stack()

stack.push(10)
stack.push(20)
stack.push(30)

print("Stack:", stack.items)
print("Top element:", stack.peek())
print("Popped element:", stack.pop())
print("Stack after pop:", stack.items)
print("Is stack empty?", stack.is_empty())

# Testing underflow
try:
    empty_stack = Stack()
    empty_stack.pop()
except IndexError as e:
    print("Error:", e)

```

4. Code Explanation

The Stack class stores its elements in a Python list. `push()` appends a new element to the end of the list, which represents the top of the stack. `pop()` and `peek()` both check `is_empty()` first and raise an `IndexError` with a clear message if the stack has no elements, preventing a stack-underflow error. `pop()` removes and returns the last item of the list, and `peek()` returns it without removing it. `is_empty()` simply checks whether the list's length is zero.

5. Output

```

Stack: [10, 20, 30]
Top element: 30
Popped element: 30
Stack after pop: [10, 20]
Is stack empty? False
Error: Stack underflow: cannot pop from an empty stack.

```

6. Complexity Analysis

Operation	Complexity
push()	O(1) amortized
pop()	O(1)
peek()	O(1)
is_empty()	O(1)

7. Screenshot

The screenshot shows a code editor with a dark theme. The editor has a menu bar (File, Edit, Selection, View, Go, Run, ...), a toolbar with icons for file operations, and a sidebar with icons for project structure, search, and other tools. The main editor area displays a Python file named `task1_stack.py` with the following code:

```
1 class Stack:
2     def __init__(self):
3         self._items = []
4     def push(self, element):
5         self._items.append(element)
6     def pop(self):
7         if self.is_empty():
8             raise IndexError("Cannot pop from an empty stack")
9         return self._items.pop()
10    def peek(self):
11        if self.is_empty():
12            raise IndexError("Cannot peek at an empty stack")
13        return self._items[-1]
14    def is_empty(self):
15        return len(self._items) == 0
16    def main():
17        stack = Stack()
18        print("Stack empty:", stack.is_empty())
19        stack.push("first")
20        stack.push("second")
21        stack.push("third")
22        print("Top after pushes:", stack.peek())
23        print("Removed:", stack.pop())
24        print("Top after pop:", stack.peek())
25        print("Removed:", stack.pop())
26        print("Removed:", stack.pop())
27        print("Stack empty after removals:", stack.is_empty())
28        for operation in (stack.pop, stack.peek):
29            try:
30                operation()
31            except IndexError as error:
32                print("Underflow handled:", error)
33    if __name__ == "__main__":
34        main()
35
```

The status bar at the bottom shows the current cursor position (Ln 32, Col 47), indentation (Spaces: 4), encoding (UTF-8), line endings (CRLF), and the Python interpreter (Python 3.14 (64-bit)).

File Edit Selection View Go Run ...

sairaj course

Problems Output Debug Console Terminal Ports

PS D:\sairaj course> & "C:\Program Files\Python314\python.exe" "d:/sairaj course/task1_stack.py"

Stack empty: True

Top after pushes: third

Removed: third

Top after pop: second

Removed: second

Removed: first

Stack empty after removals: True

Underflow handled: Cannot pop from an empty stack

Underflow handled: Cannot peek at an empty stack

PS D:\sairaj course>

Ln 32, Col 47 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live Antigra... Settings

Chat

Python stack class implementation

Start Over

Act as an experienced Python developer.

Create a Python Stack data structure using a class.

Requirements:

1. Create a class named Stack.

2. Implement these methods:

push(element)

pop()

peek()

is_empty()

3. Store stack elements using an appropriate Python data structure.

4. Handle stack underflow properly when pop() or peek() is called on an empty stack.

5. Raise an appropriate exception with a clear error message.

6. Add clear docstrings to the class and every method.

7. Add comments explaining important parts of the code.

8. Include a small demonstration program that tests all methods.

9. Explain the time complexity of each operation.

10. Keep the implementation beginner-friendly.

+ task1_stack.py

Describe what to build

+ Agent Auto Balance

Local Default permissions

Ln 84, Col 1 Spaces: 4 UTF-8 Python Python 3.14 (64-bit) Go Live Antigra... Settings

Task 2

Linear Search vs Binary Search

1.Objective

To implement Linear Search and Binary Search in Python, test both on inputs of increasing size, and compare their correctness and execution time.

2.AI Prompt Used

Act as an experienced Python algorithms developer.

Create a Python program implementing and comparing Linear Search and Binary Search.

Requirements:

1. Implement:

```
linear_search(arr, target)
binary_search(arr, target)
```

2. Add clear docstrings explaining:

- How Linear Search works
- How Binary Search works
- Their time complexity
- Their space complexity

3. Linear Search should work on an unsorted list.

4. Binary Search should work on a sorted list.

5. Return the index of the target if found.

6. Return -1 if the target is not found.

7. Test both algorithms with different input sizes such as:

```
10 elements
100 elements
1000 elements
10000 elements
```

8. Compare their execution times using Python's time module.

9. Print clear test results.

10. Add comments explaining the important steps.

3. Linear Search Code

```
import time
```

```
def linear_search(arr, target):
    """Search for target sequentially.
```

```
    Linear Search check each element from left to right.
```

TimeComplexity:

Best: $O(1)$

Average: $O(n)$

Worst: $O(n)$

SpaceComplexity:

$O(1)$

"""

for i in range(len(arr)):

if arr[i] == target:

return i

return -1

4. Binary Search Code

```
def binary_search(arr, target):
```

```
    """Search for target using the Binary Search algorithm.
```

```
    Binary Search repeatedly divides a sorted array into halves.
```

```
    TimeComplexity:
```

```
    Best:  $O(1)$ 
```

```
    Average:  $O(\log n)$ 
```

```
    Worst:  $O(\log n)$ 
```

```
    SpaceComplexity:
```

```
     $O(1)$ 
```

```
    Note:
```

```
    The input array must be sorted.
```

```
    """
```

```
    low = 0
```

```
    high = len(arr) - 1
```

```
    while low <= high:
```

```
        mid = (low + high) // 2
```

```
        if arr[mid] == target:
```

```
            return mid
```

```
        elif arr[mid] < target:
```

```
            low = mid + 1
```

```
        else:
```

```
            high = mid - 1
```

```
    return -1
```

```
# Testing different input sizes
```

```
sizes = [10, 100, 1000, 10000]
```

```
for size in sizes:
    data = list(range(size))
    target = size - 1

    start = time.perf_counter()
    linear_result = linear_search(data, target)
    linear_time = time.perf_counter() - start

    start = time.perf_counter()
    binary_result = binary_search(data, target)
    binary_time = time.perf_counter() - start

    print(f"\nInput size: {size}")
    print(f"Linear Search result: {linear_result}")
    print(f"Linear Search time: {linear_time:.8f} seconds")
    print(f"Binary Search result: {binary_result}")
    print(f"Binary Search time: {binary_time:.8f} seconds")
```

5. Explanation

linear_search() scans the list from left to right and compares each element with the target, returning its index the moment a match is found; if the loop finishes without a match it returns -1. It works on any list, sorted or not. binary_search() only works on a sorted list: it repeatedly compares the target with the middle element of the current search range and discards the half of the range that cannot contain the target, narrowing low and high until the element is found or the range is empty.

6. Complexity Analysis

Algorithm	Best	Average	Worst	Space
Linear Search	O(1)	O(n)	O(n)	O(1)
Binary Search	O(1)	O(log n)	O(log n)	O(1)

7. Test Results

Use your own measured times from running task2_search.py — do not invent numbers.

Input Size	Linear Search Time	Binary Search Time
10		
100		
1000		
10000		

8. Comparison

Binary Search has a worst-case time complexity of O(log n), which is asymptotically faster than Linear Search's O(n), but it only works on data that is already sorted, while Linear Search works on unsorted data too. In practice, the measured times above depend on the machine, the Python version and the specific run, so Binary Search will not necessarily look faster on every single measurement, especially for very small input sizes — the true advantage becomes clear as the input size grows.

9. Screenshot

File Edit Selection View Go Run ...

sairaj course

task2_search.py X prime_one_shot.py X

task2_search.py > run_tests

```
1 import time
2 def linear_search(arr, target):
3     for index, value in enumerate(arr):
4         if value == target:
5             return index
6     return -1
7 def binary_search(arr, target):
8     left = 0
9     right = len(arr) - 1
10    while left <= right:
11        middle = (left + right) // 2
12
13        if arr[middle] == target:
14            return middle
15        if arr[middle] < target:
16            left = middle + 1
17        else:
18            right = middle - 1
19    return -1
20 def measure_time(search_function, arr, target, repetitions=100):
21     start = time.perf_counter()
22     result = -1
23     for _ in range(repetitions):
24         result = search_function(arr, target)
25     elapsed = time.perf_counter() - start
26     return result, elapsed / repetitions
27 def run_tests():
28     input_sizes = (10, 100, 1000, 10000)
29     print("Search Algorithm Performance Comparison")
30     print("-" * 72)
31     print(f"{'Size':>8} {'Linear Index':>15} {'Linear Time':>15} "
32           f"{'Binary Index':>15} {'Binary Time':>15}")
33     print("-" * 72)
34     for size in input_sizes:
35         sorted_values = list(range(size))
36         unsorted_values = sorted_values[::2] + sorted_values[1::2]
```

Ln 47, Col 73 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live Antigravity - Settings

Explorer

Test Explorer

sairaj course

__pycache__

.vscode

project settings

.env

.gitignore

CRC4.py

Lab_11_Data_Structu...

prime_one_shot.py

python gemini_test.py

Realtime.py

task2_search.py

users.db

Timeline

Chat

Python linear and binary search imple...

Act as an experienced Python algorithms developer....

behavior to preserve. I'm adding a self-contained script with deterministic test data, a small timing helper, and separate unsorted/sorted inputs so the comparison exercises each algorithm under its required contract.

Edited task2_search.py +101 -1

The implementation is in place. I'm running the script now as the focused check; this will validate both return values and the printed timing table across all four requested input sizes.

1 confirmation pending

Run pwsh command?

python task2_search.py

Runs task2_search.py — executes the search comparison script.

Allow Skip

+ task2_search.py

Describe what to build

+ Agent A... Balance ↕ ⌵

Local Default permissions

File Edit Selection View Go Run ...

sairaj course

task2_search.py X prime_one_shot.py X

task2_search.py > run_tests

```
27 def run_tests():
34     for size in input_sizes:
35         sorted_values = list(range(size))
36         unsorted_values = sorted_values[::2] + sorted_values[1::2]
37         target = size - 1
38
39         linear_index, linear_time = measure_time(
40             linear_search, unsorted_values, target
41         )
42         binary_index, binary_time = measure_time(
43             binary_search, sorted_values, target
44         )
45         print(f"{'size':>8} {'linear_index':>15} "
46               f"{'linear_time * 1_000_000':>12.3f} us "
47               f"{'binary_index':>15} {'binary_time * 1_000_000':>12.3f} us")
48 if __name__ == "__main__":
49     run_tests()
50
```

Ln 47, Col 73 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live Antigravity - Settings

Explorer

Test Explorer

sairaj course

__pycache__

.vscode

project settings

.env

.gitignore

CRC4.py

Lab_11_Data_Structu...

prime_one_shot.py

python gemini_test.py

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task2_search.py

users.db

Timeline

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Edited task2_search.py +101 -1

The implementation is in place. I'm running the script now as the focused check; this will validate both return values and the printed timing table across all four requested input sizes.

1 confirmation pending

Run pwsh command?

python task2_search.py

Runs task2_search.py — executes the search comparison script.

Allow Skip

+ task2_search.py

Describe what to build

+ Agent A... Balance ↕ ⌵

Local Default permissions

FileEditSelectionViewGoRun...sairaj course

ProblemsOutputDebug ConsoleTerminalPorts

Python: tas...

PS D:\sairaj course> & "C:\Program Files\Python314\python.exe" "d:/sairaj course/task2_search.py"

Search Algorithm Performance Comparison

Size	Linear Index	Linear Time	Binary Index	Binary Time
10	9	0.427 us	9	0.361 us
100	99	1.765 us	99	0.449 us
1000	999	21.782 us	999	0.771 us
10000	9999	269.487 us	9999	1.051 us

PS D:\sairaj course>

Ln 47, Col 73Spaces: 4UTF-8CRLFPythonPython 3.14 (64-bit)Go LiveAntigravity - Settings

FileEditSelectionViewGoRun...sairaj course

Chat

Sessions

Python linear and binary search implementation

Act as an experienced Python algorithms developer....

Create a Python program implementing and comparing Linear Search and Binary Search.

Requirements:

1. Implement:
linear_search(arr, target)
binary_search(arr, target)

2. Add clear docstrings explaining:

- How Linear Search works
- How Binary Search works
- Their time complexity
- Their space complexity

3. Linear Search should work on an unsorted list.

4. Binary Search should work on a sorted list.

5. Return the index of the target if found.

6. Return -1 if the target is not found.

7. Test both algorithms with different input sizes such as:
10 elements
100 elements
1000 elements
10000 elements

8. Compare their execution times using Python's time module.

9. Print clear test results.

+ task2_search.py

Describe what to build

+ Agent Auto Balance

LocalDefault permissions

Ln 47, Col 73Spaces: 4UTF-8CRLFPythonPython 3.14 (64-bit)Go LiveAntigravity - Settings

Task 3

Test Driven Development – Calculator

1. Objective

To build a simple Calculator using Test Driven Development: writing the tests first, watching them fail, then writing the implementation until all tests pass.

2. AI Prompt for Test Generation

Act as an experienced Python Test Driven Development developer.

Create pytest unit tests for a simple calculator.

Requirements:

1. Create tests for addition.
2. Create tests for subtraction.
3. Test positive numbers.
4. Test negative numbers.
5. Test zero.
6. The implementation does not exist yet because this is TDD.
7. Put all tests in a separate file named test_calculator.py.
8. Use clear test function names.
9. Add comments explaining what each test verifies.

3. Test Code (test_calculator.py)

```
from task3_calculator import add, subtract
```

```
def test_add_positive_numbers():  
    """Test addition of two positive numbers."""  
    assert add(10, 5) == 15
```

```
def test_add_negative_numbers():  
    """Test addition involving negative numbers."""  
    assert add(-10, -5) == -15
```

```
def test_add_zero():  
    """Test addition with zero."""  
    assert add(10, 0) == 10
```

```
def test_subtract_positive_numbers():  
    """Test subtraction of two positive numbers."""  
    assert subtract(10, 5) == 5
```

```
def test_subtract_negative_numbers():  
    """Test subtraction involving negative numbers."""
```

```
assert subtract(-10, -5) == -5
```

```
def test_subtract_zero():  
    """Test subtraction with zero."""  
    assert subtract(10, 0) == 10
```

4. Initial Test Execution

Command run: `pytest` (executed before `task3_calculator.py` existed).

5. Failure Observed

Because `task3_calculator.py` did not exist yet, the import `"from task3_calculator import add, subtract"` failed, so `pytest` reported a collection error (`ModuleNotFoundError`) and all tests failed. This confirms the tests were written and executed before any implementation existed, as required by TDD.

6. AI Prompt for Implementation

The `pytest` tests have already been written for a calculator.

Now implement the required calculator functions so that all existing tests pass.

Requirements:

1. Create `task3_calculator.py`.
2. Implement `add(a, b)`.
3. Implement `subtract(a, b)`.
4. Add clear docstrings.
5. Add simple comments.
6. Do not modify the test file.
7. Keep the implementation simple and readable.

7. Calculator Implementation (`task3_calculator.py`)

```
def add(a, b):  
    """Return the sum of two numbers."""  
    return a + b  
  
def subtract(a, b):  
    """Return the difference between two numbers."""  
    return a - b
```

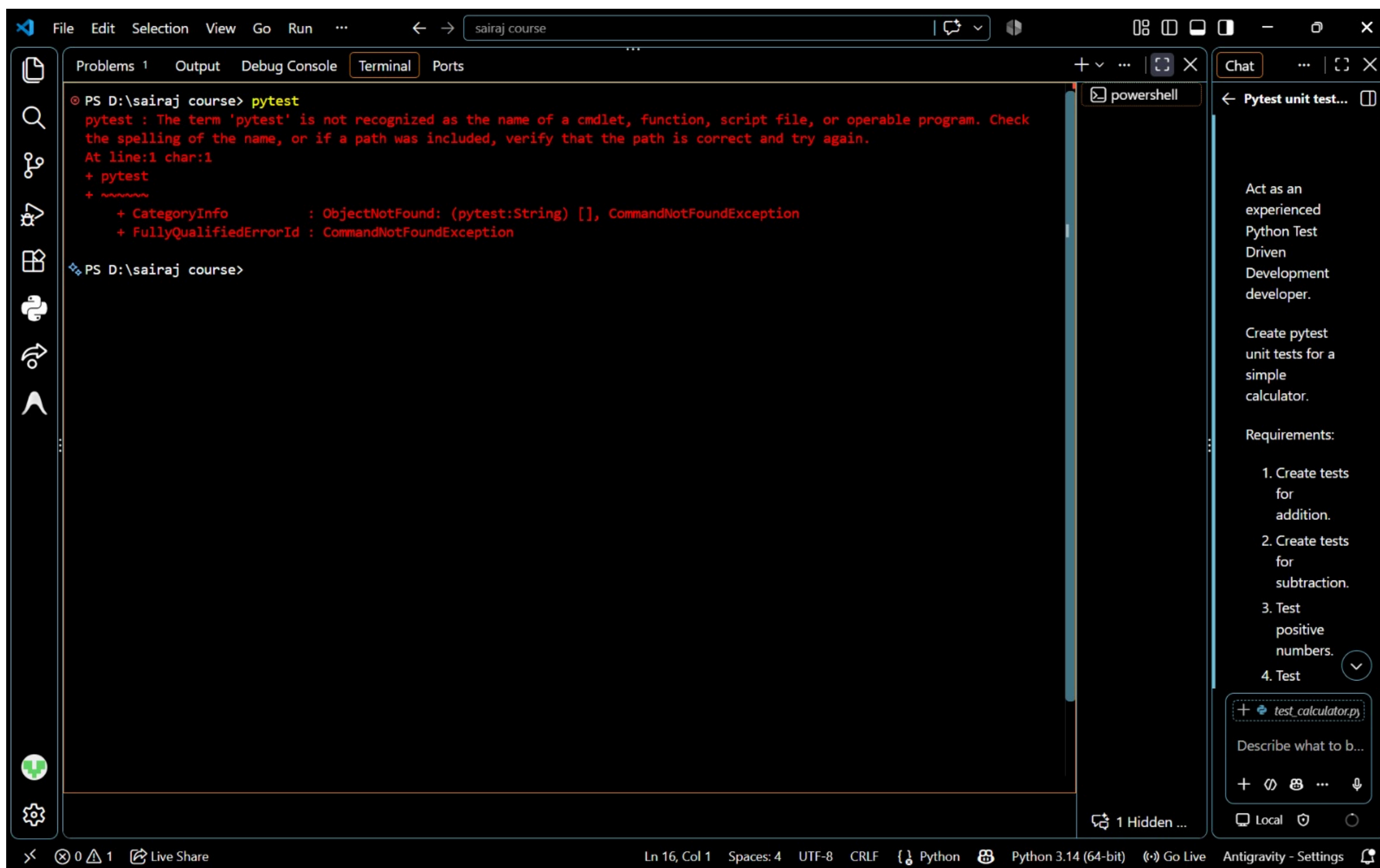
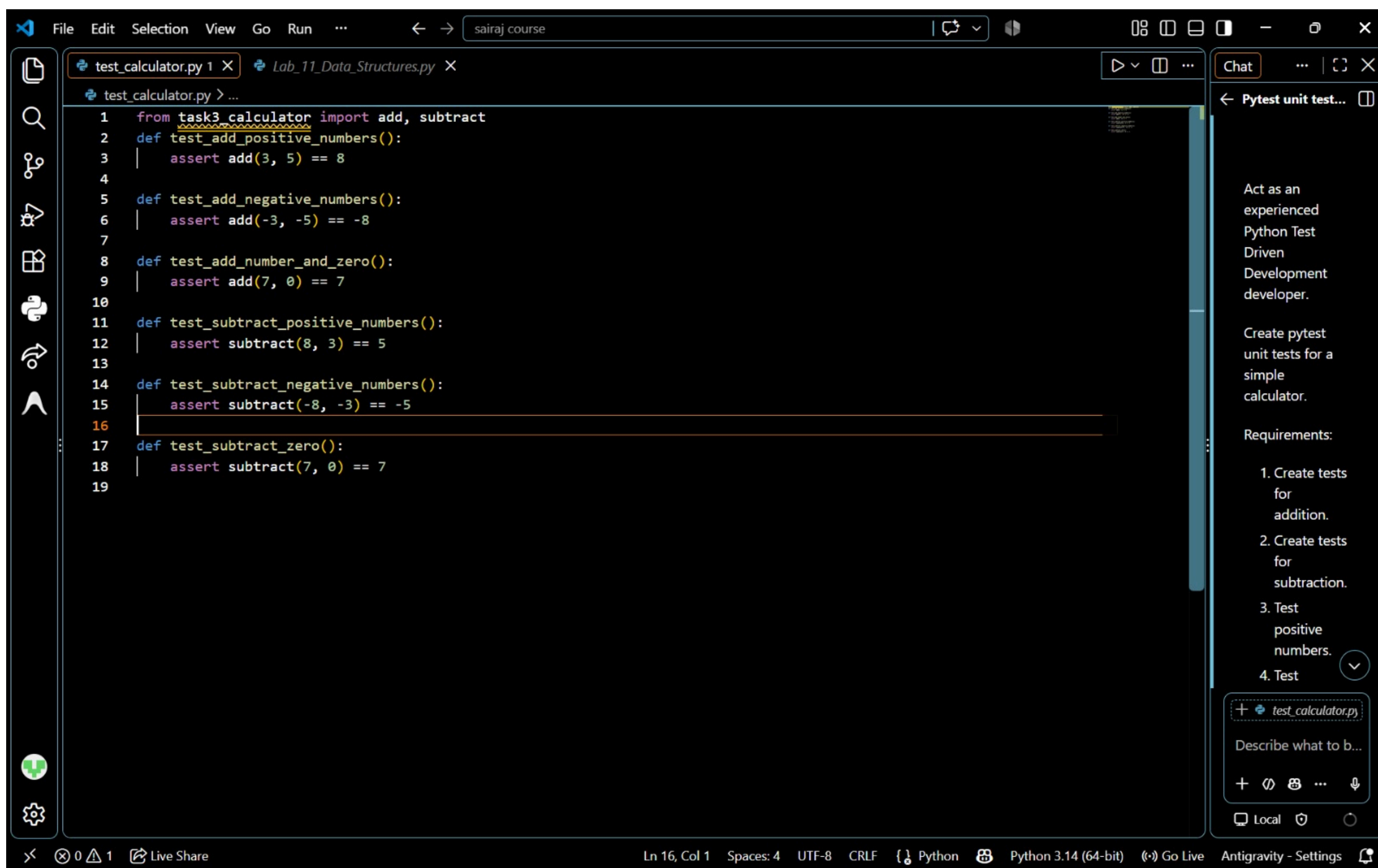
8. Final Test Execution

Command run: `pytest` (executed after `task3_calculator.py` was created).

All 6 tests passed:

6 passed

10. Screenshots



FileEditSelectionViewGoRun...sairaj course

Chat

Pytest unit tests for calculator

Start Over

Act as an experienced Python Test Driven Development developer.

Create pytest unit tests for a simple calculator.

Requirements:

1. Create tests for addition.

2. Create tests for subtraction.

3. Test positive numbers.

4. Test negative numbers.

5. Test zero.

6. The implementation does not exist yet because this is TDD.

7. Put all tests in a separate file named test_calculator.py.

8. Use clear test function names.

9. Add comments explaining what each test verifies.

3:13 PM

Completed 11 steps in 1m 8s

Created test_calculator.py with six pytest tests covering:

Addition and subtraction

Positive and negative numbers

+ test_calculator.py

Describe what to build

+ Agent Auto Balance

Local Default permissions

Sessions

New Session

Today3

Pytest unit tests for calculator

4 mins ago

Python linear and binary search impleme...

+101 -1 · 12 mins ago

Python stack class implementation

+84 -1 · 18 mins ago

Archived6

FileEditSelectionViewGoRun...sairaj course

Chat

Pytest unit tests for calculator

The pytest tests have already been written for a calculator.

Now implement the required calculator functions so that all existing tests pass.

Requirements:

1. Create task3_calculator.py.

2. Implement add(a, b).

3. Implement subtract(a, b).

4. Add clear docstrings.

5. Add simple comments.

6. Do not modify the test file.

7. Keep the implementation simple and readable.

Completed 6 steps in 33s

Created task3_calculator.py with:

add(a, b)

subtract(a, b)

Clear docstrings

Simple explanatory comments

All existing tests pass: 6 passed.

+ test_calculator.py

Describe what to build

+ Agent Auto Balance

Local Default permissions

Sessions

New Session

Today3

Pytest unit tests for calculator

+13 -1 · 10 mins ago

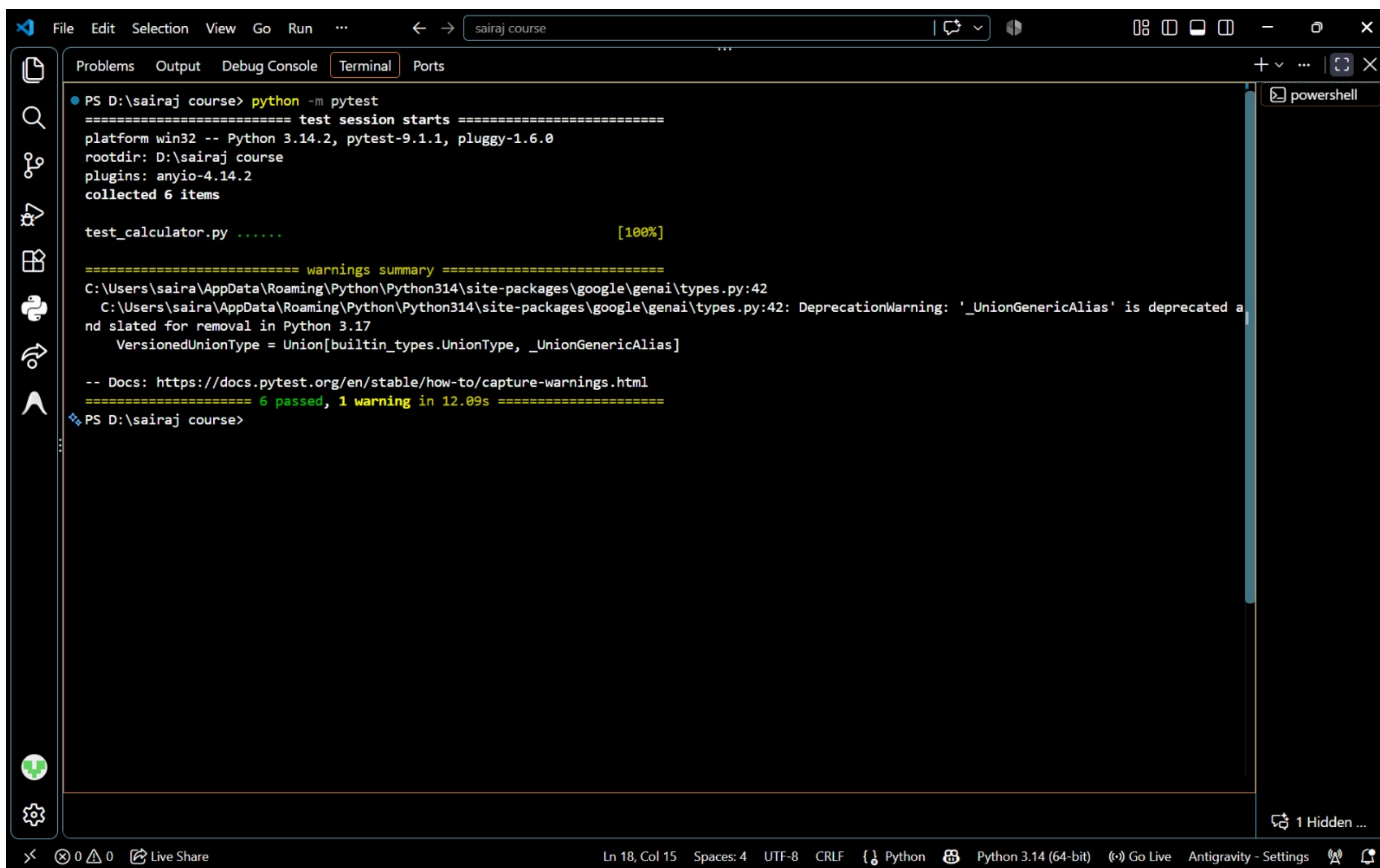
Python linear and binary search impleme...

+101 -1 · 18 mins ago

Python stack class implementation

+84 -1 · 24 mins ago

Archived6



Task 4

Queue Implementation with AI Assistance

1. Objective

To design and implement a Queue data structure in Python with the help of AI assistance, supporting enqueue, dequeue, front and is_empty operations with FIFO behavior, and with proper handling of queue overflow and underflow.

2. AIPromptUsed

Act as an experienced Python data structures developer.

Create a Queue data structure using a Python class.

Requirements:

1. Create a class named Queue.
2. Implement:
 - enqueue(element)
 - dequeue()
 - front()
 - is_empty()
3. Use FIFO (First In First Out) behavior.
4. Handle queue underflow when dequeue() or front() is called on an empty queue.
5. Handle queue overflow using a configurable maximum queue size.
6. Raise clear exceptions for overflow and underflow.
7. Add docstrings to the class and every method.
8. Add comments explaining important operations.
9. Include a demonstration program testing normal operations, underflow and overflow.
10. Explain the time and space complexity.

3. Python Code

```
class Queue:
    """A Queue implementation using FIFO behavior."""

    def __init__(self, max_size=5):
        """Initialize an empty queue with a maximum capacity."""
        self.items = []
        self.max_size = max_size

    def enqueue(self, element):
        """Add an element to the rear of the queue.

        Raises:
            OverflowError: If the queue has reached maximum capacity.
        """
        if len(self.items) >= self.max_size:
            raise OverflowError("Queue overflow: queue is full.")

        self.items.append(element)
```

```
def dequeue(self):
    """Remove and return the front element.

    Raises:
        IndexError: If the queue is empty.
    """
    if self.is_empty():
        raise IndexError("Queue underflow: queue is empty.")

    return self.items.pop(0)

def front(self):
    """Return the front element without removing it.

    Raises:
        IndexError: If the queue is empty.
    """
    if self.is_empty():
        raise IndexError("Queue underflow: queue is empty.")

    return self.items[0]

def is_empty(self):
    """Return True if the queue is empty."""
    return len(self.items) == 0
```

Demonstration

```
queue = Queue(max_size=3)
```

```
queue.enqueue(10)
```

```
queue.enqueue(20)
```

```
queue.enqueue(30)
```

```
print("Queue:", queue.items)
```

```
print("Front element:", queue.front())
```

```
print("Dequeued:", queue.dequeue())
```

```
print("Queue after dequeue:", queue.items)
```

```
print("Is queue empty?", queue.is_empty())
```

Testing overflow

```
try:
```

```
    queue.enqueue(40)
```

```
    queue.enqueue(50)
```

```
    queue.enqueue(60)
```

```
except OverflowError as e:
```

```
    print("Error:", e)
```

Testing underflow

```
try:
```

```
    empty_queue = Queue()
```

```
    empty_queue.dequeue()
```

```
except IndexError as e:
    print("Error:", e)
```

4. Code Explanation

The Queue class stores elements in a Python list and follows First In First Out (FIFO) order. enqueue() adds a new element to the rear (end) of the list, after checking it has not reached max_size. dequeue() and front() both check is_empty() first; dequeue() removes and returns the element at index 0 (the front of the queue), and front() returns that same element without removing it.

5. Overflow Handling

The queue is created with a configurable max_size (3 in the demonstration). When enqueue() is called and the queue already holds max_size elements, it raises an OverflowError with the message "Queue overflow: queue is full." instead of silently accepting the new element.

6. Underflow Handling

When dequeue() or front() is called on a queue with no elements, is_empty() returns True and an IndexError is raised with the message "Queue underflow: queue is empty.", preventing an attempt to read from an empty queue.

7. Output

Expected output (values will vary slightly depending on when the queue reaches max_size):

```
Queue: [10, 20, 30]
Front element: 10
Dequeued: 10
Queue after dequeue: [20, 30]
Is queue empty? False
Error: Queue overflow: queue is full.
Error: Queue underflow: queue is empty.
```

8. Complexity Analysis

Operation	Complexity	Note
enqueue()	O(1) amortized	Appends to the end of the list
dequeue()	O(n)	pop(0) shifts every remaining element
front()	O(1)	Direct index access
is_empty()	O(1)	Length check

Note: dequeue() is O(n) here because Python lists shift all remaining elements when the first item is removed with pop(0). A collections.deque would give O(1) dequeue() if that trade-off matters for your report.

9. Screenshot

File Edit Selection View Go ... ← → sairaj course

task4.queue.py X

task4.queue.py > Queue > enqueue

```
1 from collections import deque
2 class Queue:
3     def __init__(self, max_size):
4         if isinstance(max_size, bool) or not isinstance(max_size, int):
5             raise TypeError("max_size must be a positive integer")
6         if max_size <= 0:
7             raise ValueError("max_size must be a positive integer")
8         self._items = deque()
9         self._max_size = max_size
10    def enqueue(self, element):
11        if self.is_full():
12            raise OverflowError(f"Queue overflow: maximum size of {self._max_size} reached")
13        self._items.append(element)
14    def dequeue(self):
15        if self.is_empty():
16            raise IndexError("Queue underflow: cannot dequeue from an empty queue")
17        return self._items.popleft()
18    def front(self):
19        if self.is_empty():
20            raise IndexError("Queue underflow: cannot read front of an empty queue")
21        return self._items[0]
22    def is_empty(self):
23        return not self._items
24    def is_full(self):
25        return len(self._items) == self._max_size
26 def demonstrate_queue():
27     queue = Queue(max_size=3)
28     print("Queue is empty:", queue.is_empty())
29     queue.enqueue("first")
30     queue.enqueue("second")
31     queue.enqueue("third")
32     print("Front element:", queue.front())
33     print("Dequeued elements:", queue.dequeue(), queue.dequeue())
34     queue.enqueue("fourth")
35     print("Next front element:", queue.front())
36     print("Remaining elements:", queue.dequeue(), queue.dequeue())
37     print("Queue is empty:", queue.is_empty())
38     try:
39         queue.dequeue()
40     except IndexError as error:
41         print("Underflow handled:", error)
42     try:
43         queue.front()
44     except IndexError as error:
```

0 0 0 Live Share

Ln 12, Col 33 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live Antigravity - Settings

File Edit Selection View Go ... ← → sairaj course

task4.queue.py X

task4.queue.py > Queue > enqueue

```
26 def demonstrate_queue():
38     try:
39         queue.dequeue()
40     except IndexError as error:
41         print("Underflow handled:", error)
42     try:
43         queue.front()
44     except IndexError as error:
45         print("Front underflow handled:", error)
46     queue.enqueue("one")
47     queue.enqueue("two")
48     queue.enqueue("three")
49     try:
50         queue.enqueue("four")
51     except OverflowError as error:
52         print("Overflow handled:", error)
53 if __name__ == "__main__":
54     demonstrate_queue()
55
56
57
58
```

0 0 0 Live Share

Ln 12, Col 33 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live Antigravity - Settings

File Edit Selection View Go Run ...

sairaj course

Python: tas...

Explorer

Test Explorer

sairaj course

__pycache__

.pytest_cache

.vscode

project settings

.env

.gitignore

CRC4.py

Lab_11_Data_Structu...

prime_one_shot.py

python gemini_test.py

Realtime.py

task4_queue.py

users.db

Timeline

Problems

Output

Debug Console

Terminal

Ports

PS D:\sairaj course> & "C:\Program Files\Python314\python.exe" "d:/sairaj course/task4_queue.py"

Queue is empty: True

Front element: first

Dequeued elements: first second

Next front element: third

Remaining elements: third fourth

Queue is empty: True

Underflow handled: Queue underflow: cannot dequeue from an empty queue

Front underflow handled: Queue underflow: cannot read front of an empty queue

Overflow handled: Queue overflow: maximum size of 3 reached

PS D:\sairaj course>

Ln 23, Col 31

Spaces: 4

UTF-8

CRLF

Python

Python 3.14 (64-bit)

Go Live

Antigravity - Settings

File Edit Selection View Go Run ...

sairaj course

Python: tas...

Chat

Python Queue class implementation

Start Over

Act as an experienced Python data structures developer.

Create a Queue data structure using a Python class.

Requirements:

1. Create a class named Queue.

2. Implement:

enqueue(element)

dequeue()

front()

is_empty()

3. Use FIFO (First In First Out) behavior.

4. Handle queue underflow when dequeue() or front() is called on an empty queue.

5. Handle queue overflow using a configurable maximum queue size.

6. Raise clear exceptions for overflow and underflow.

7. Add docstrings to the class and every method.

8. Include a demonstration program testing normal operations, underflow and overflow.

9. Explain the time and space complexity.

Completed 16 steps in 1m 1s

> 1 file changed +5 -0

Keep Undo

+ task4_queue.py

Describe what to build

+ Agent Auto Balance

Local Default permissions

Sessions

New Session

Today

4

Python Queue class implementation

+5 -0 · 2 mins ago

Pytest unit tests for calculator

+13 -1 · 1 hr ago

Python linear and binary search impleme...

+101 -1 · 2 hrs ago

Python stack class implementation

+84 -1 · 2 hrs ago

Archived

6

Ln 104, Col 1

Spaces: 4

UTF-8

Python

Python 3.14 (64-bit)

Go Live

Antigravity - Settings

Task 5

Bubble Sort vs Selection Sort

1.Objective

To implement Bubble Sort and Selection Sort in Python without using built-in sort functions, and to compare the two algorithms on the same input list.

2.AI Prompt Used

```
Act as an experienced Python algorithms developer.

Create a Python program implementing Bubble Sort and Selection Sort.

Requirements:

1. Implement:
    bubble_sort(arr)
    selection_sort(arr)

2. Both functions should sort a list in ascending order.

3. Do not use Python's built-in sorted() or sort() functions.

4. Add a clear docstring to each function.

5. Docstrings must mention:
    - Best-case time complexity
    - Average-case time complexity
    - Worst-case time complexity
    - Space complexity

6. Add comments explaining each important step of both algorithms.

7. Use copies of the input list so the original data can be preserved.

8. Test both algorithms with:
    [64, 25, 12, 22, 11]

9. Print the original list and both sorted results.

10. Keep the implementation beginner-friendly.
```

3. Bubble Sort Code

```
def bubble_sort(arr):
    """Sort a list using Bubble Sort.

    Time Complexity:
        Best: O(n)
        Average: O(n^2)
        Worst: O(n^2)

    Space Complexity:
```

```

    O(1) auxiliary space for the in-place algorithm.
    """
    n = len(arr)

    # Compare adjacent elements repeatedly.
    for i in range(n):
        swapped = False

        for j in range(0, n - i - 1):

            # Swap if elements are in the wrong order.
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
                swapped = True

        # Stop early if no swaps occurred.
        if not swapped:
            break

    return arr

```

4. Selection Sort Code

```

def selection_sort(arr):
    """Sort a list using Selection Sort.

    Time Complexity:
    Best: O(n^2)
    Average: O(n^2)
    Worst: O(n^2)

    Space Complexity:
    O(1) auxiliary space for the in-place algorithm.
    """
    n = len(arr)

    # Find the smallest element for each position.
    for i in range(n):

        min_index = i

        # Search for the smallest remaining element.
        for j in range(i + 1, n):
            if arr[j] < arr[min_index]:
                min_index = j

        # Place the smallest element at position i.
        arr[i], arr[min_index] = arr[min_index], arr[i]

    return arr

```

```
# Testing
original = [64, 25, 12, 22, 11]

bubble_data = original.copy()
selection_data = original.copy()

print("Original list:", original)
print("BubbleSort:", bubble_sort(bubble_data))
print("SelectionSort:", selection_sort(selection_data))
```

5. Step-by-Step Explanation

Bubble Sort repeatedly walks through the list, compares each pair of adjacent elements, and swaps them if they are in the wrong order; after each full pass the next-largest element "bubbles" into its correct position at the end, and a swapped flag lets the algorithm stop early once a pass makes no swaps. Selection Sort instead finds the minimum element in the unsorted remainder of the list on each pass and swaps it into the current position, growing the sorted portion of the list from the front one element at a time.

6. Complexity Analysis

Algorithm	Best	Average	Worst	Auxiliary Space
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(1)$

7. Test Output

```
Original list: [64, 25, 12, 22, 11]
Bubble Sort: [11, 12, 22, 25, 64]
Selection Sort: [11, 12, 22, 25, 64]
```

8. Comparison

Both algorithms sort the same list [64, 25, 12, 22, 11] into [11, 12, 22, 25, 64]. Bubble Sort can finish in $O(n)$ time in the best case (an already-sorted list, thanks to the swapped flag), while Selection Sort always takes $O(n^2)$ comparisons regardless of the input order, since it scans the remaining unsorted portion on every pass. Both use only $O(1)$ extra space, since they sort the list in place.

9. Screenshot

The image shows a VS Code editor interface with a dark theme. The Explorer sidebar on the left shows a project named 'sairaj course' with several files, including 'task5_sorting.py' which is currently open in the editor. The editor displays a Python script implementing bubble sort and selection sort. The script includes docstrings for both functions and a test case at the bottom. The Chat sidebar on the right shows a conversation with an AI assistant. The chat messages include instructions to act as an experienced Python algorithms developer, test both algorithms with a specific list, print the original and sorted results, and keep the implementation beginner-friendly. The assistant has completed 9 steps in 31 seconds and implemented the 'task5_sorting.py' file. It lists the included features: bubble sort and selection sort functions, complexity details, beginner-friendly comments, input-list preservation using '.copy()', and a required sample test and output. The assistant confirms that both algorithms produce the correct sorted list [11, 12, 22, 25, 64] and that no errors were found.

```
1 def bubble_sort(arr):
2     result = arr.copy()
3     for pass_number in range(len(result) - 1):
4         swapped = False
5         for index in range(len(result) - 1 - pass_number):
6             if result[index] > result[index + 1]:
7                 result[index], result[index + 1] = result[index + 1], result[index]
8                 swapped = True
9         if not swapped:
10            break
11    return result
12
13 def selection_sort(arr):
14     result = arr.copy()
15
16     for current_index in range(len(result) - 1):
17         smallest_index = current_index
18         for index in range(current_index + 1, len(result)):
19             if result[index] < result[smallest_index]:
20                 smallest_index = index
21
22         result[current_index], result[smallest_index] = (
23             result[smallest_index],
24             result[current_index],
25         )
26
27    return result
28
29 numbers = [64, 25, 12, 22, 11]
30 print("Original list:", numbers)
31 print("Bubble Sort result:", bubble_sort(numbers))
32 print("Selection Sort result:", selection_sort(numbers))
```

Chat

← Bubble sort and selection sort implementation

Act as an experienced Python algorithms developer....

8. Test both algorithms with:
[64, 25, 12, 22, 11]

9. Print the original list and both sorted results.

10. Keep the implementation beginner-friendly.

Completed 9 steps in 31s

Implemented `task5_sorting.py`.

It includes:

- `bubble_sort(arr)`
- `selection_sort(arr)`
- Complexity details in both docstrings
- Beginner-friendly comments
- Input-list preservation using `.copy()`
- Required sample test and output

Both algorithms produce `[11, 12, 22, 25, 64]`. No errors found.

+ `task5_sorting.py`

Describe what to build

+ Agent Auto Balance

Local Default permissions

The image shows a screenshot of a Visual Studio Code (VS Code) editor interface. The main window displays a Python script named `task5_sorting.py` in the `Terminal` tab. The script's output is visible, showing the execution of a bubble sort and a selection sort on a list of numbers.

The `Terminal` output shows:

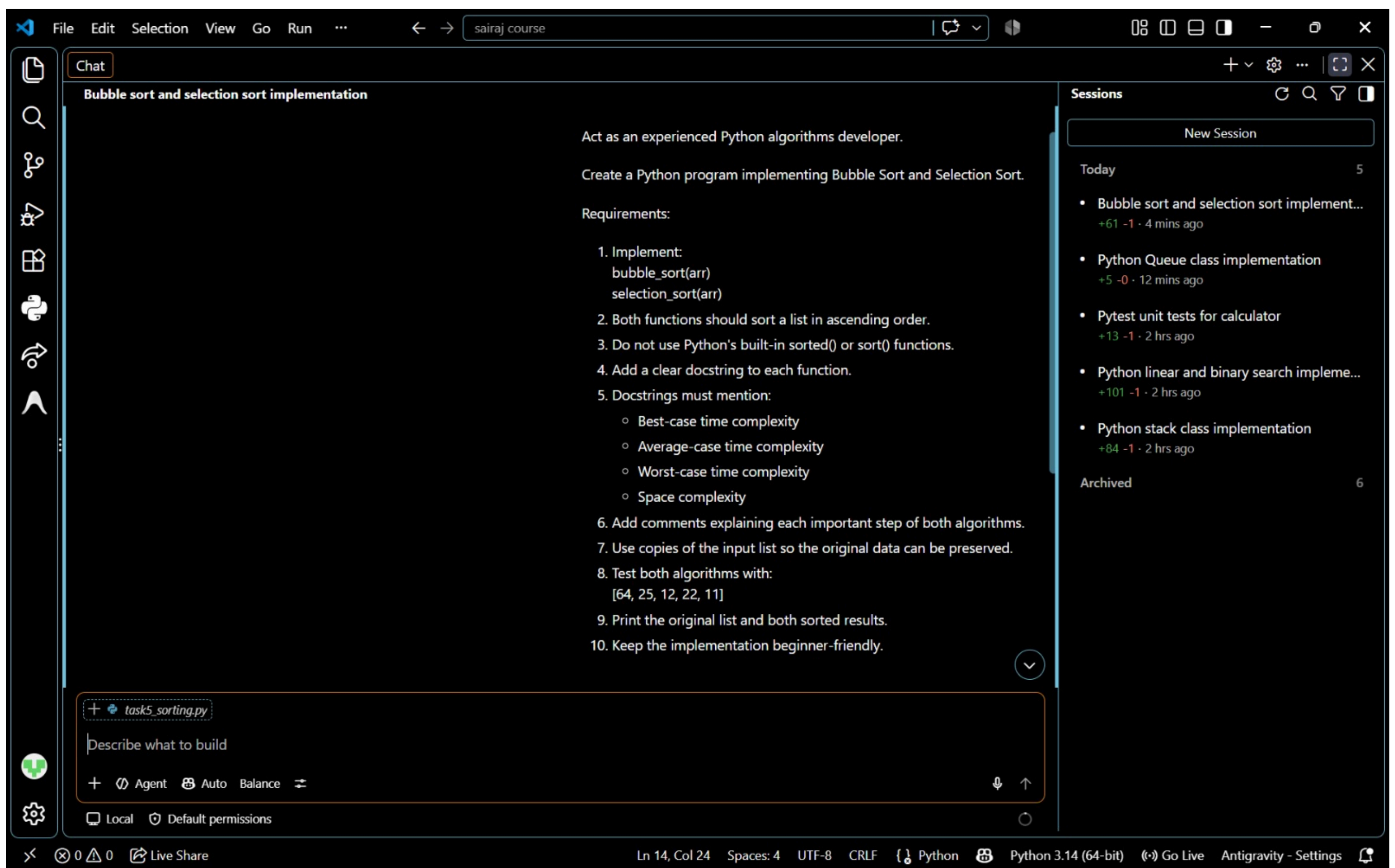
```
PS D:\sairaj course> & "C:\Program Files\Python314\python.exe" "d:\sairaj course\course\task5_sorting.py"
Original list: [64, 25, 12, 22, 11]
Bubble Sort result: [11, 12, 22, 25, 64]
Selection Sort result: [11, 12, 22, 25, 64]
PS D:\sairaj course>
```

The `Chat` window on the right side of the editor shows a conversation with an AI assistant. The assistant has completed 9 steps in 31s and has implemented the bubble sort and selection sort algorithms. The assistant's response includes a list of features and a sample test and output.

The `Chat` window content includes:

- Completed 9 steps in 31s
- Implemented `task5_sorting.py`.
- It includes:
 - `bubble_sort(arr)`
 - `selection_sort(arr)`
 - Complexity details in both docstrings
 - Beginner-friendly comments
 - Input-list preservation using `.copy()`
 - Required sample test and output
- Both algorithms produce `[11, 12, 22, 25, 64]`. No errors found.

The `Chat` window also shows a section for describing what to build, with a text input field and a button to generate code.



Conclusion

AI assistance helped me to build each data structure/algorithm, about TDD that taught me about writing tests before code, and how the complexity analysis of each approach compared in practice.]